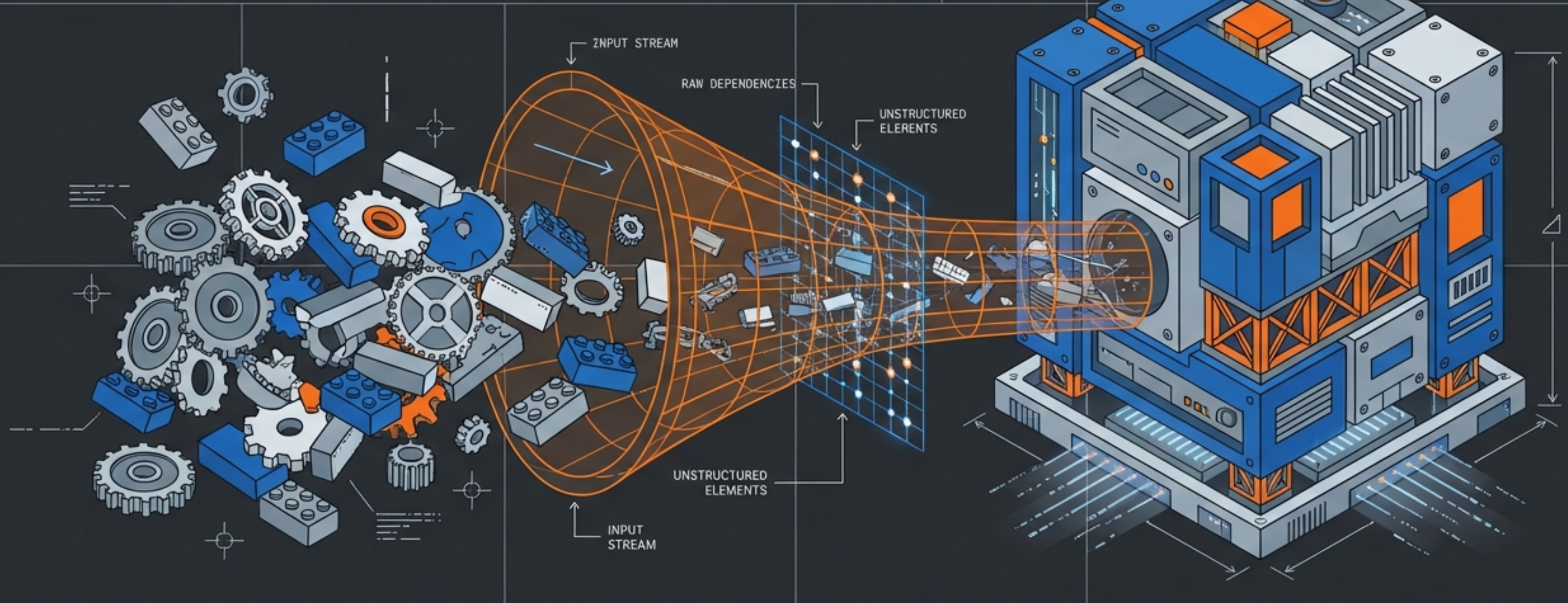


APACHE MAVEN

THE ARCHITECTURE OF BUILD AUTOMATION

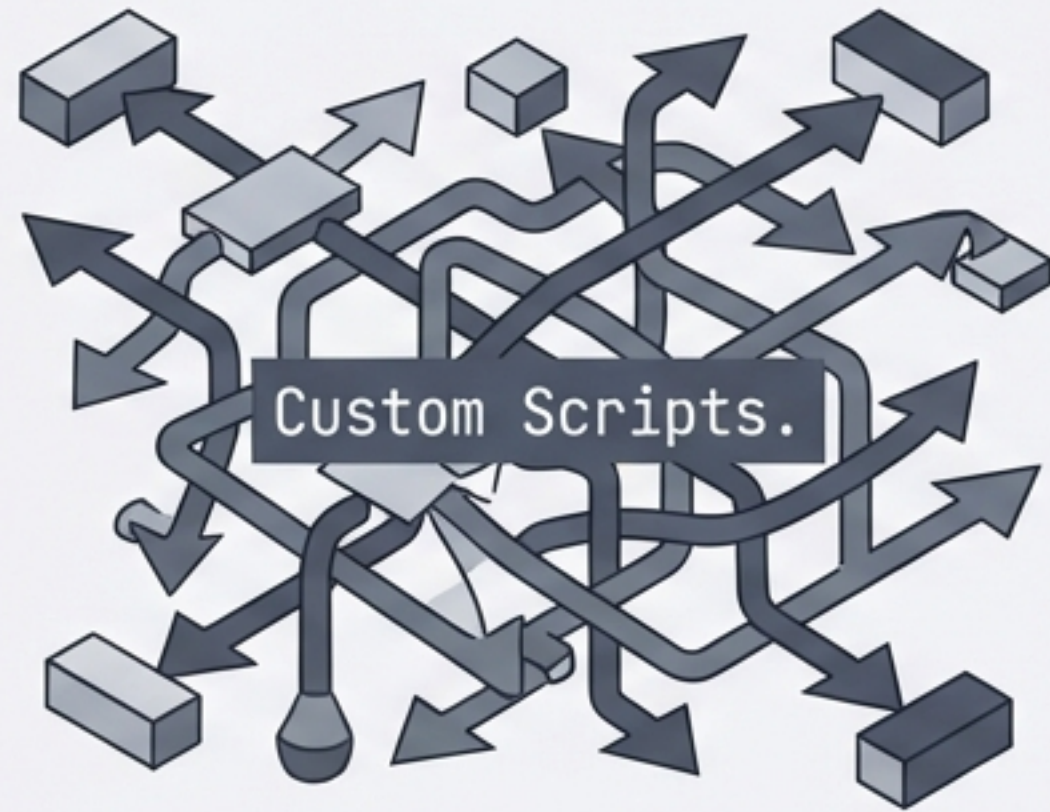


Mastering the philosophy of Convention over Configuration.

Maven is more than a build tool; it is a project management tool that provides a shared mental model for Java projects. By standardizing the build lifecycle and directory structure, It allows developers to Jump between projects without learning new setup procedures.

PHILOSOPHY: CONVENTION OVER CONFIGURATION

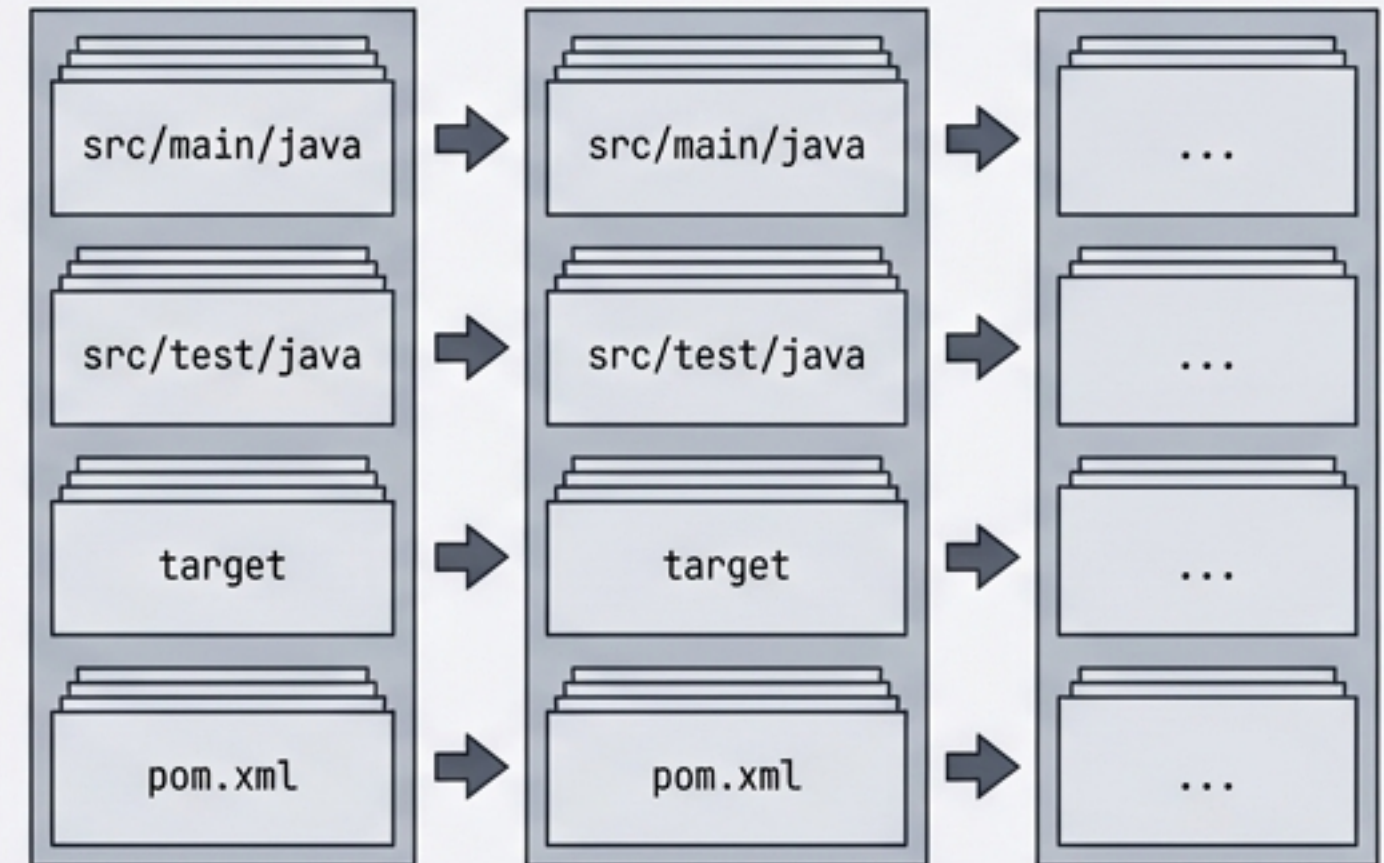
Imperative: Scripting the Process (Ant)



You explicitly tell the tool *where* source code lives and *how* to compile it. Every project has a unique, non-standard layout.

Result: High cognitive load.

Declarative: Declaring the Intent (Maven)



Maven is opinionated. It assumes a standard behavior. If you follow the rules, the build "just works".

Result: Focus on code, not plumbing.

THE BLUEPRINT: PROJECT OBJECT MODEL (POM)

```
1 <project>
2   <modelVersion>4.0.0</modelVersion>
3
4   <groupId>com.infinite.skills</groupId>
5   <artifactId>maven-hello-world</artifactId>
6   <version>1.0.0</version>
7
8   <dependencies>...</dependencies>
9 </project>
```

GROUP ID (The Organization)

The unique identifier of the organization, usually the domain name in reverse.
Example: com.infinite.skills

ARTIFACT ID (The Project)

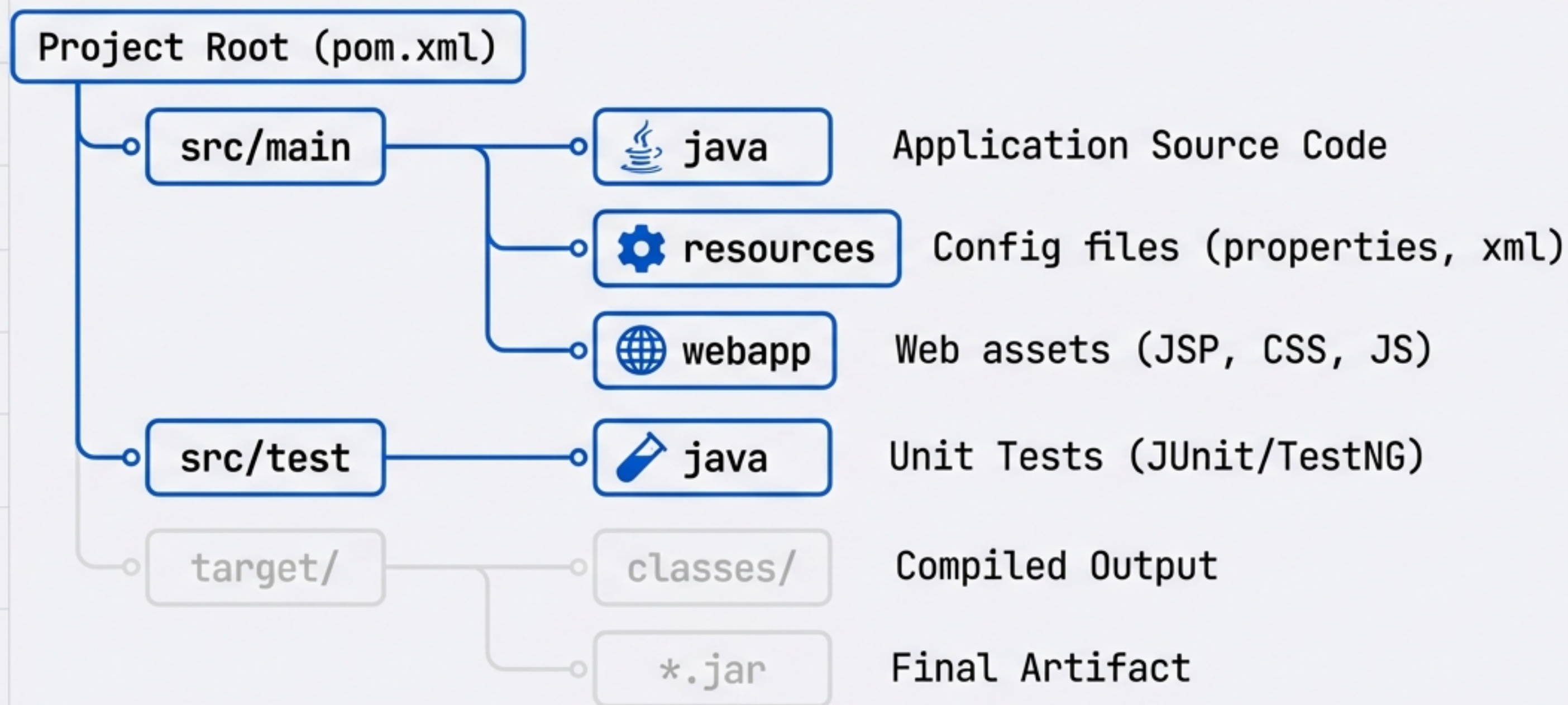
The name of the specific project or module.
Example: maven-hello-world

VERSION (The Time)

The timestamp or release tag of the build.
Example: 1.0.0

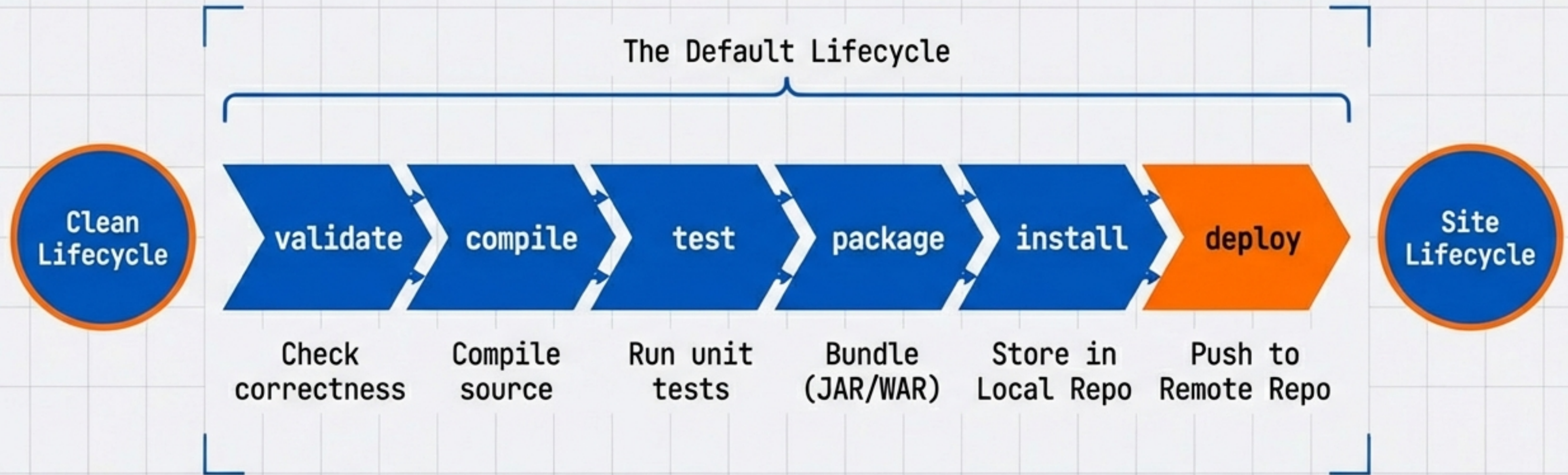
The G.A.V. Coordinate System: The address system of the Maven universe.

THE STANDARD DIRECTORY LAYOUT



Adhering to this layout eliminates the need to configure source paths. Maven knows where to look.

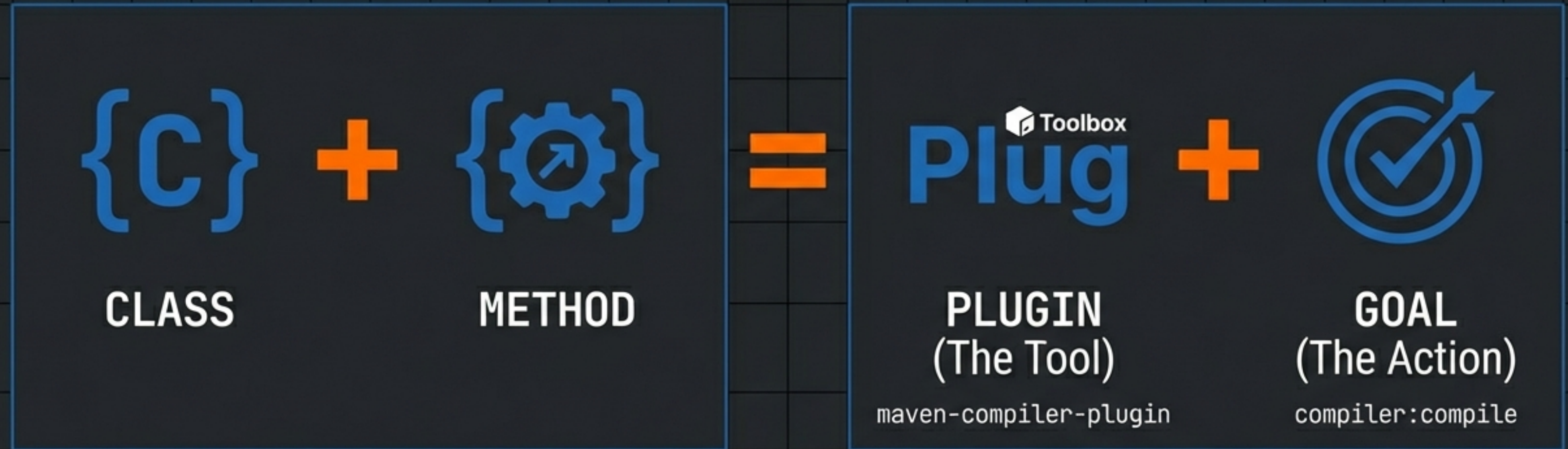
THE BUILD LIFECYCLE: A SEQUENTIAL PROCESS



Executing a later phase (e.g., install) automatically executes all preceding phases (validate, compile, test, package).

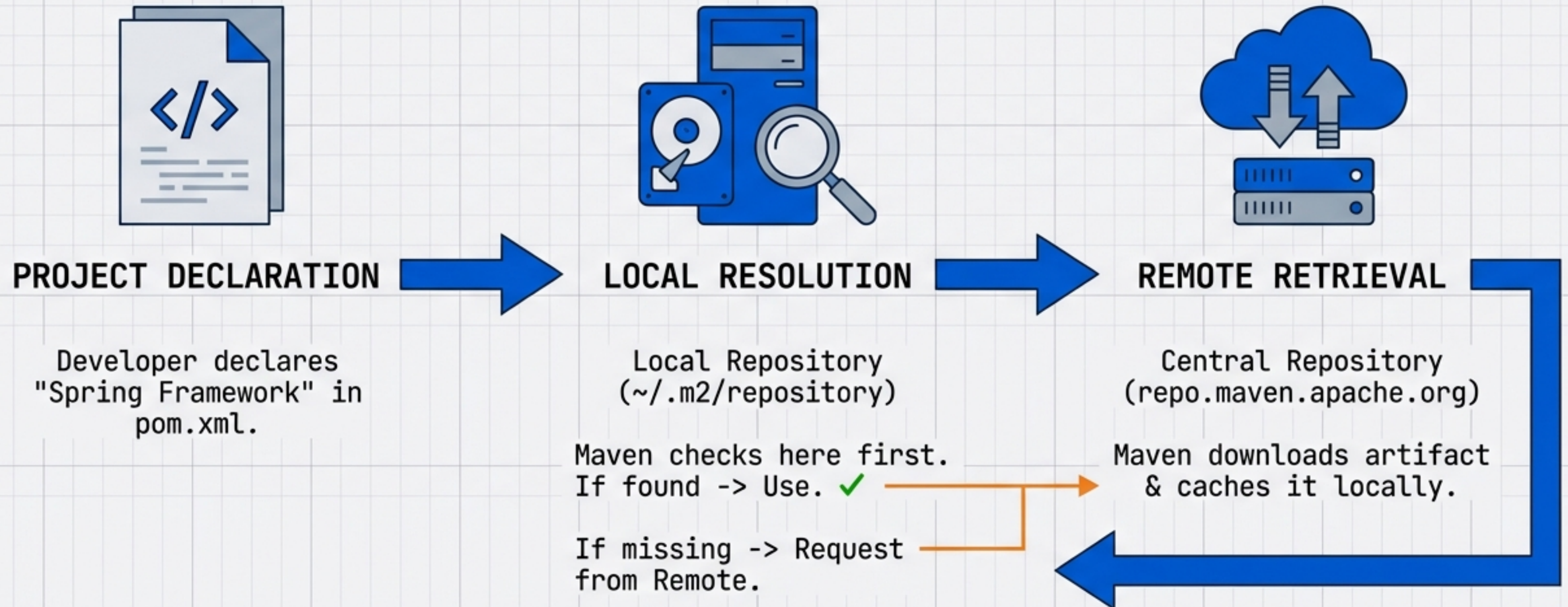
THE MECHANISM: PLUGINS & GOALS

Maven is a framework for executing plugins.



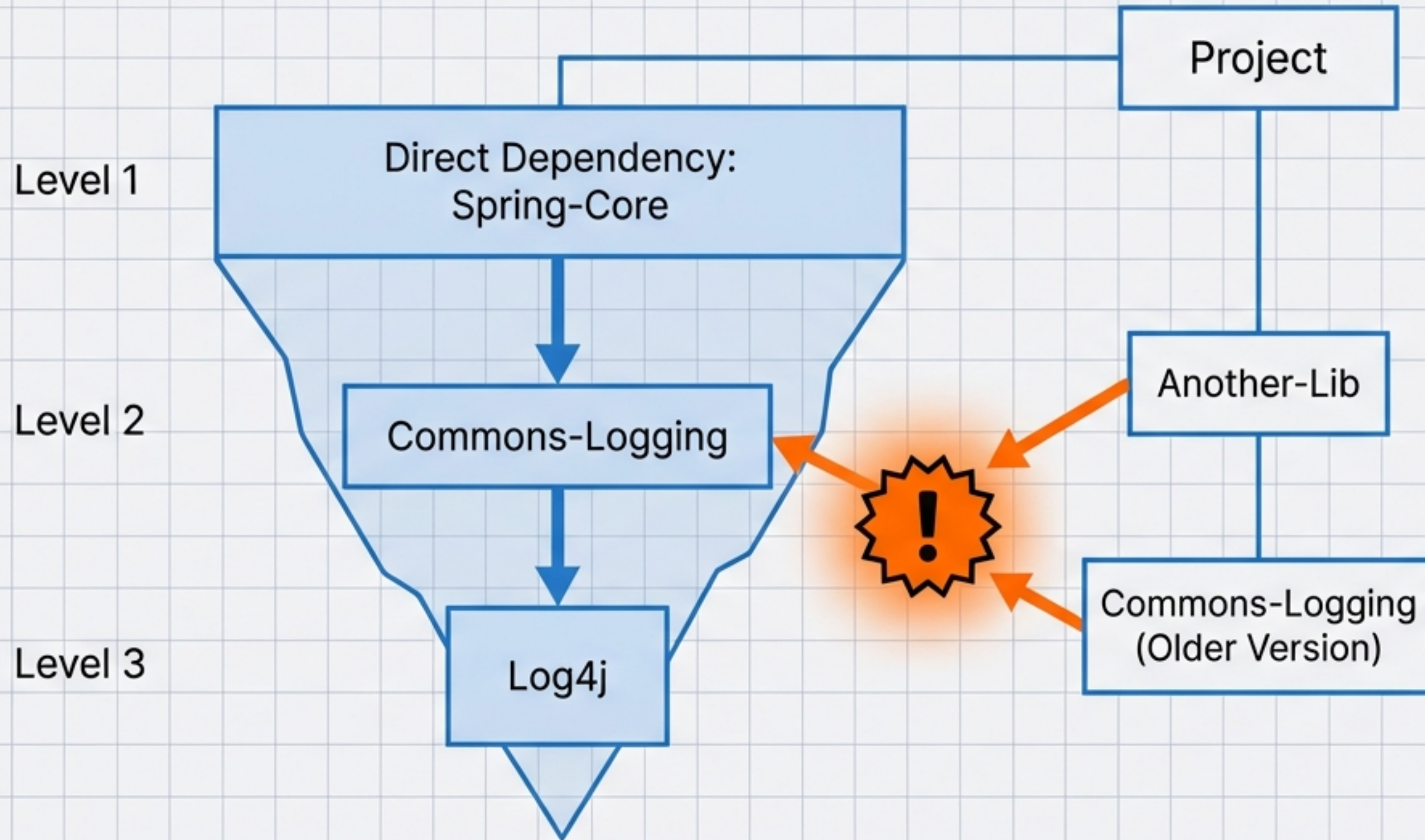
Binding: Goals are bound to Lifecycle phases. When the 'compile' phase triggers, it invokes the 'compiler:compile' goal.

DEPENDENCY MANAGEMENT: THE ECOSYSTEM



KEY INSIGHT: Eliminates manual JAR downloads. The POM is the single source of truth for all libraries.

TRANSITIVE DEPENDENCIES & CONFLICTS



Conflict Resolution Rules:

- 1. Nearest Definition Wins:** Maven picks the version closest to your project in the tree.
- 2. Exclusions:** Use the `<exclusion>` tag in POM to manually block unwanted transitives (preventing 'JAR Hell').

```
<exclusion>  
  <groupId>commons-logging</groupId>  
  <artifactId>commons-logging</artifactId>  
</exclusion>
```


DEPENDENCY SCOPES: CONTEXT MATTERS

COMPILE (Default)

Available in: Build, Test, Run.

Commons-Lang, Spring-Core



TEST

Available in: Test
Compilation/Execution only.

JUnit, Mockito



PROVIDED

Needed for compilation, provided by
container at runtime.

Servlet API (Tomcat provides this)



RUNTIME

Not needed for compile, needed to run.

JDBC Driver Implementation



MULTI-MODULE PROJECTS & INHERITANCE

PARENT POM (pom.xml)

Defines: Compiler Version, Shared Plugins, Dependency Management.

Packaging: pom

MODULE A (Service Layer)

Inherits Parent settings.

Defines specific dependencies.

MODULE B (Web Layer)

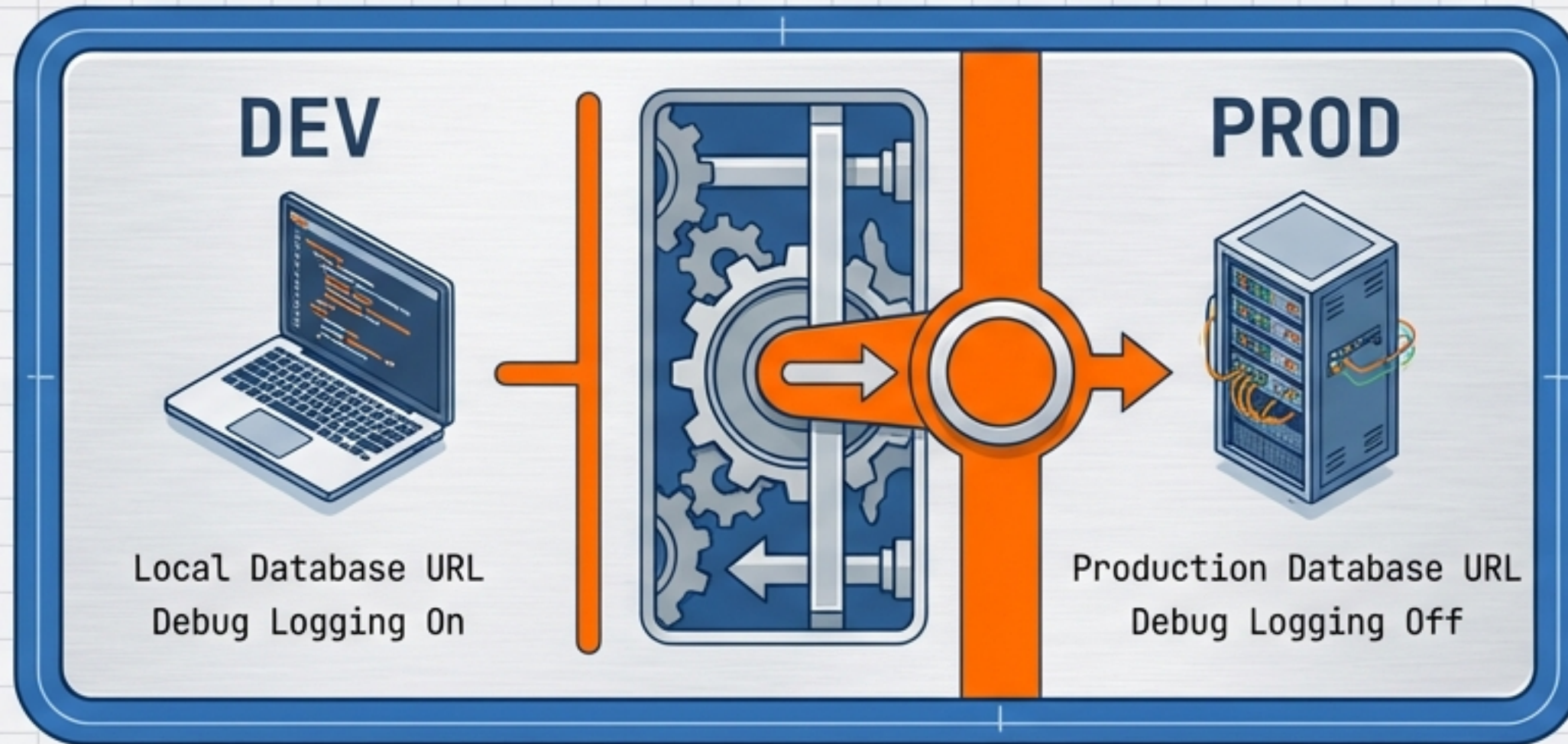
Inherits Parent settings.

Depends on Module A.

Aggregation vs Inheritance

- **Inheritance:** Sharing configuration (Don't Repeat Yourself).
- **Aggregation:** Building the Parent automatically builds all Modules in the correct order.

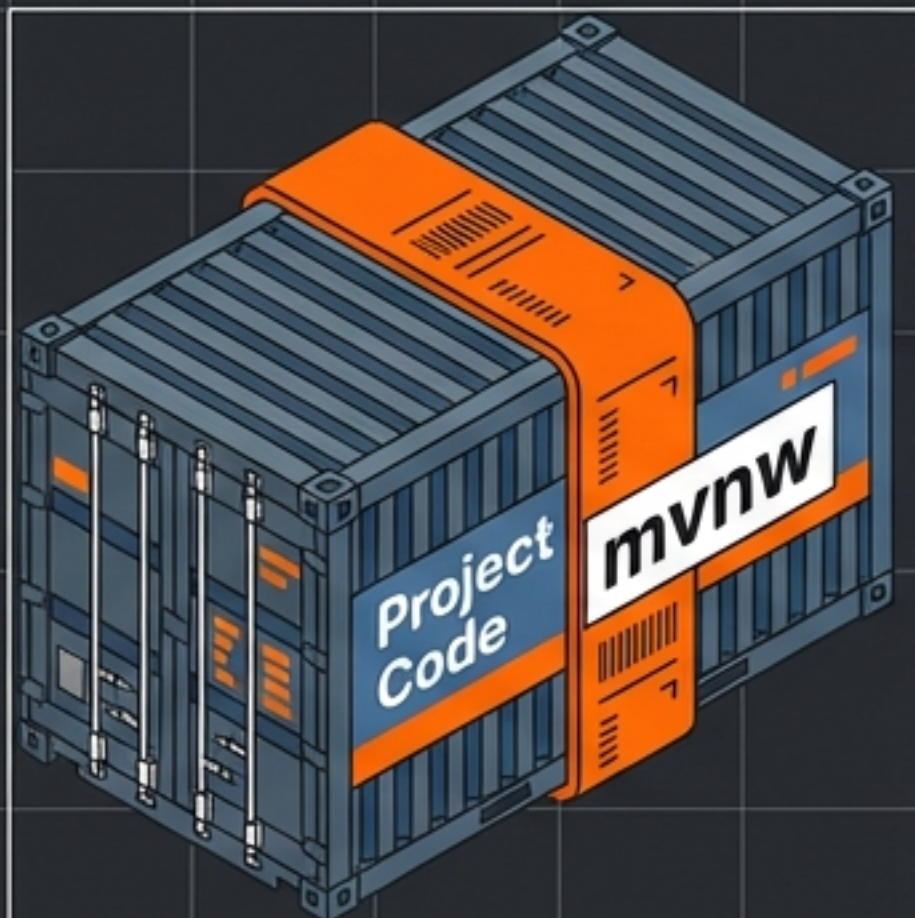
PROFILES: ADAPTABILITY FOR ENVIRONMENTS



Mechanism: Use `<profiles>` in `pom.xml` to override configurations.

Activation: Trigger via command line (`-P production`), JDK version, or OS.

THE MAVEN WRAPPER (mvnw)



Ensures the build works even if Maven is not installed on the developer's machine. Locks the Maven version for consistency.

ESSENTIAL COMMANDS

\$ <code>./mvnw clean install</code>	(Builds and tests the project)
\$ <code>./mvnw dependency:tree</code>	(Visualizes dependency hierarchy)
\$ <code>./mvnw archetype:generate</code>	(Scaffolds new project)
\$ <code>./mvnw site</code>	(Generates documentation)

Maven brings order to chaos. Transition from scripting builds to architecting them.